Chuyu Yan

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Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML, CSS
Frontend: React, React Router, Context API, Vite, Tailwind CSS
Backend & Frameworks: Flask, REST APIs, SQLAlchemy ORM, JWT, Selenium, OpenCV, Tesseract OCR, PyPDF2, PyMuPDF (fitz), pdfplumber, PIL/Pillow
Cloud & Infrastructure: Azure AI Foundry, Azure AI Search, Azure Document Intelligence, Azure Blob Storage, Azure Cosmos DB, Docker, Linux, Gunicorn, Nginx, Git, Alibaba Cloud
Databases: SQLite
AI & Search: FAISS, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Microsoft Presidio, Semantic Search, Vector Search, LLM Integration, Code Embeddings, AST Parsing

Experience

AI Agent Developer — **May 2026 – Aug 2026**
Arthur Health — *Toronto, Canada*

- Co-architected and built an end-to-end **multi-agent platform** that automates clinical report quality assurance by transforming unstructured reports into structured evaluations and RAG-grounded recommendations.
- Designed an identity-anchored **PHI redaction** system that combines **Microsoft Presidio** with header-derived identity resolution, fuzzy name matching, and ownership-aware rules to accurately remove patient identifiers while preserving clinician and clinic information.
- Designed a rate-limit-aware parallel execution pipeline using **ThreadPoolExecutor**, reducing recommendation latency by up to **5**$\times$ while ensuring deterministic output despite concurrent execution.
- Designed a **table reconstruction algorithm** that restored structured report layouts from **Azure Document Intelligence** output, enabling reliable downstream evaluation while preserving backward compatibility.

Software Engineering Intern — **Sept 2025 – Dec 2025**
Dingke Medical Technology — *Suzhou, China*

- Designed and deployed a hybrid code search platform built on **Flask REST APIs**, integrating **ripgrep** and **FAISS (IndexFlatIP)** retrieval to support **1,000+** concurrent users.
- Built scalable **FAISS** indexing with incremental updates across **2,000+** repositories and **50k+** code chunks, combining vector similarity with keyword search to improve retrieval accuracy by **60%**.
- Designed multi-tenant repository storage using **SQLite**, **SQLAlchemy ORM**, **JWT**, and **RBAC**, securing **20+** REST endpoints for **500+** registered users.
- Implemented multi-provider **LLM orchestration** with caching and exponential backoff retry logic, reducing API costs by **40%**.
- Containerized services with **Docker** and deployed to an **Alibaba Cloud Linux VM** using **Gunicorn** and **Nginx**, achieving **99.9%** uptime.

Software R&D Intern — **Jan 2025 – Apr 2025**
Kangyu Medical Devices — *Shijiazhuang, China*

- Designed a **document detection pipeline** using **OpenCV** for **800+** heterogeneous, low-quality medical documents, enabling reliable downstream OCR extraction.
- Engineered an automated batch processing pipeline integrating **Selenium**, **PDF extraction**, and **Tesseract OCR**, reducing manual processing time by **99%**.

Projects

Code Search & Developer Tooling Platform | *React 18, TypeScript, Vite, Flask, FAISS, Tree-sitter, Docker*

- Developed a **VS Code extension** with Webview-based chat, enabling in-editor semantic code search and AI-assisted developer workflows.
- Implemented incremental **Tree-sitter AST** indexing with file monitoring to support low-latency repository updates.
- Built a real-time developer dashboard using **React**, **TypeScript**, and **Server-Sent Events (SSE)** for interactive code analysis.

Hospital Integration Automation | *Python, Selenium WebDriver, Multi-threading, JSON Logging*

- Developed **Python** and **Selenium WebDriver**-based automation to migrate structured patient data from hospital intranet systems to third-party case management platforms, reducing processing time by **73%** (**45s** $\rightarrow$ **12s** per record).
- Engineered a data transformation pipeline handling **20–30** heterogeneous fields with schema validation and type conversion, successfully migrating **800+** patient records with **99.5%** data integrity.
- Implemented configurable multi-threading with exponential backoff retry, validation, and rollback mechanisms, increasing throughput from **30** to **200** records/day while supporting production deployment at a **Grade III Class A hospital**.

Education

University of Waterloo — **Expected Apr 2029**
Bachelor of Applied Science in Computer Engineering — *Waterloo, ON*

- **Relevant Coursework**: Data Structures & Algorithms, Digital Computers, Digital Circuits, Linear Circuits, Linear Algebra